Appendix A
Simulation Pseudocode for an ABM of Brokerage in Matching Markets

1 State, Notation, and Matching Function

Blue notation, equations, and algorithm lines are diagnostics-only measures. They are recorded for $\mathcal{M}$ or downstream figures, exploration scripts, and analyses; they do not enter transition, choice, learning, or matching rules.

For any matrix Z, $Z_{\cdot\ell}$ denotes column ℓ and $Z_{\cdot L}$ denotes the submatrix with ordered column-index set L. For a vector z, z_L denotes the corresponding subvector.

Table 1: Cross-routine symbols and dimensions.

Symbol	Dimension/domain	Definition
Core		
N	scalar	Number of agents, with agent set $[N] = \{1, \ldots, N\}$.
ϑ	parameter vector	Simulation parameter vector.
seed	integer	Random seed.
S^t	state tuple	Mutable simulation state at end of period t.
T	integer	Number of simulation periods.
$\mathcal{M} = [m^1; \ldots; m^T]$	$\mathbb{R}^{T\times m}$	Measure table; row $m^t = \mathcal{M}_{t\cdot} \in \mathbb{R}^m$.
Parameters		
d	scalar	Agent type dimension.
k_G	scalar	Watts-Strogatz graph degree.
ω	scalar	Satisfaction EWMA weight.
L	learning function	NN (default) or Ridge.
S_b	broker service	Full (default), assessment-only, or access-only.
λ_a, λ_b	positive scalars	Agent and broker Ridge slope penalties; calibrated experiment values are $\lambda_a = 0.003$ and $\lambda_b = 0.001$.
a_b	broker Ridge variant	Pair, size-matched, single-principal, or additive.
$E_{\text{init},a}, E_{\text{init},b}$	scalars	Agent and broker initial training steps; both are 100 in the reporting simulations.
$E_{\text{steps},a}, E_{\text{steps},b}$	scalars	Agent and broker minimum training steps per fitting call; both are 50 in the reporting simulations.
$\eta_{\text{lr},a}, \eta_{\text{lr},b}$	scalars	Agent and broker Adam learning rates; reporting values are 0.003 and 0.03, respectively.
W_p	scalar	Training-window horizon, in periods.
M	scalar	Per-call training observation cap.
Agent types, networks, and output		
$G^t = (V, \mathcal{E}_G^t)$	graph	Undirected graph, $V = [N] \cup \{N+1\}$; node $N+1$ is the broker.

Symbol	Dimension/domain	Definition
$X^t = [x_1^t, \ldots, x_N^t]$	$\mathbb{R}^{d\times N}$	Agent type matrix; $x_i^t = X_{\cdot i}^t \in \mathbb{R}^d$.
γ	$[0,1] \to \mathbb{R}^d$	Agent type curve for initial and entrant draws.
q_{ij}^t	scalar	Realized output for match (i,j).
Match-output function		
Q	scalar	Output offset.
ρ	scalar	General-quality share of the two components' summed marginal variances.
δ	scalar	Strength of the pair boundary in match-specific complementarity.
σ_ε	scalar	Output-noise standard deviation.
c	$\mathbb{R}^d$	Ideal agent type.
A	$\mathbb{R}^{d\times d}$	Symmetric positive definite complementarity matrix.
B	$\mathbb{R}^{d\times d}$	Symmetric boundary matrix.
Calibration and costs		
$\bar{q}_{\mathrm{cal}}$	scalar	Calibration mean.
r	scalar	Reservation output.
ϕ	scalar	Broker fee.
c_s	scalar	Self-search cost.
Learning and predictions		
d_b	integer	Broker symmetric pair-feature dimension, $d + d(d+1)/2$.
$\psi_b(x_i, x_j)$	$\mathbb{R}^{d_b}$	Symmetric broker pair feature vector.
$\mathcal{H}_i^t = (\Xi_i^t, y_i^t, n_i^t)$	$\Xi_i^t \in \mathbb{R}^{d\times n_i^t}$, $y_i^t \in \mathbb{R}^{n_i^t}$	Agent i history; observation ℓ is partner agent type $\Xi_{i,\cdot\ell}^t$ and output $y_{i\ell}^t$.
$\mathcal{H}_b^t = (X_{b,1}^t, X_{b,2}^t, y_b^t, n_b^t)$	$X_{b,1}^t, X_{b,2}^t \in \mathbb{R}^{d\times n_b^t}$, $y_b^t \in \mathbb{R}^{n_b^t}$	Broker history; observation ℓ stores the two party types and output $y_{b\ell}^t$.
Θ_i^t	predictor parameters	Agent i NN weights or Ridge coefficients.
Θ_b^t	predictor parameters	Broker NN weights or Ridge coefficients.
$\mathcal{O}_i^t, \mathcal{O}_b^t$	Adam states	Persistent NN optimizer moments and step counters; unused by Ridge.
Δ_i^t, Δ_b^t	nonnegative integers	Observations added since the learner's previous training call.
$\mathbf{p}_i^t, \mathbf{p}_b^t$	integer sequences	Cumulative history counts at the end of learner initialization (period 0) and each later completed period.
$v_b^t(i,j)$	scalar	Symmetric broker prediction for unordered pair $\{i,j\}$.
$a_{i\to j}^t$	scalar	Agent i's assessment of candidate j.
$w_{j\leftarrow i}^t$	scalar	Receiver acceptance score.
$\bar{q}_{i\cdot}^t$	vector	Agent i's partner-indexed empirical means.
$C_{i\cdot}^t$	vector	Agent i's partner-indexed counts.
Period state and market objects		
s_i^t	$\mathbb{R}^2$	Satisfaction scores for self and broker channels.
χ_i^t	binary	Indicator that agent i has ever used the broker.
M_i^t	set	Agent i current-period counterparties.
z_i^t	channel	Channel chosen by demander i.
d_i^t	integer	Current-period demand.
D^t	subset of $[N]$	Current broker clients, $\{i : d_i^t > 0, z_i^t = \text{broker}\}$.
$\mathcal{A}_b^t$	subset of $[N]$	Broker access set, $\text{Roster}^t \cup D^t$.
Roster^t	subset of $[N]$	Standing roster.

Symbol	Dimension/domain	Definition
rep_b^t	scalar	Broker reputation.
D_+^t	subset of $[N]$	Broker clients with positive residual offer quota.
$\mathcal{D}^t$	list	Demand records (i, z_i^t, d_i^t) with $d_i^t > 0$.
$\mathcal{R}^t$	list	Accepted records $(i, j, z, q, \hat{q}, o_1, o_2)$.
rng^t	RNG state	State of the simulation random-number stream.

The mutable simulation state at the end of period t is

$$S^t = \begin{pmatrix} \vartheta, \mathsf{rng}^t, \gamma, c, A, B, \bar{q}_{\mathrm{cal}}, r, \phi, c_s, X^t, G^t, \\ \{\mathcal{H}_i^t, \Theta_i^t, \mathcal{O}_i^t, \Delta_i^t, \mathbf{p}_i^t, \bar{q}_{i\cdot}^t, C_{i\cdot}^t, s_i^t, \chi_i^t, M_i^t\}_{i=1}^N, \\ \mathcal{H}_b^t, \Theta_b^t, \mathcal{O}_b^t, \Delta_b^t, \mathbf{p}_b^t, \mathrm{Roster}^t, D^t, \mathrm{rep}_b^t \end{pmatrix}.$$

This tuple includes fixed realized model objects and all variables needed for later transitions. Implementation-only scratch buffers and diagnostic-only counters are omitted.

For the fixed reference moments, define

$$U_{ij} = \frac{x_i^\top c + x_j^\top c}{2}, \qquad V_{ij} = x_i^\top A x_j, \qquad H_{ij} = \mathrm{sign}(x_i^\top B x_j) V_{ij},$$

$$\widetilde{U}_{ij} = \frac{U_{ij} - \mu_U}{\sigma_U}, \qquad \widetilde{W}_{ij}(\delta) = \frac{V_{ij} + \delta H_{ij} - (\mu_V + \delta \mu_H)}{\sqrt{\sigma_V^2 + 2\delta \,\mathrm{Cov}(V, H) + \delta^2 \sigma_H^2}}.$$

With $\kappa(\delta) = \mathrm{Corr}(U, W(\delta))$, the systematic output component is

$$f(x_i, x_j) = \frac{\sqrt{\rho}\, \widetilde{U}_{ij} + \sqrt{1-\rho}\, \widetilde{W}_{ij}(\delta)}{\sqrt{1 + 2\sqrt{\rho(1-\rho)}\, \kappa(\delta)}}.$$

Expected match output is $Q + f(x_i, x_j)$. A completed match produces realized output

$$q_{ij} = Q + f(x_i, x_j) + \varepsilon_{ij}, \qquad \varepsilon_{ij} \sim \mathcal{N}(0, \sigma_\varepsilon^2).$$

2 Top-Level Simulation

Algorithm 1 Brokerage Simulation

Require: Parameters ϑ and seed.
Ensure: Final state S^T and measure table $\mathcal{M} \in \mathbb{R}^{T \times m}$.

1: $S^0 \leftarrow \textsc{Initialize}(\vartheta, \mathrm{seed})$
2: **for** $t = 1, \ldots, T$ **do**
3: $\quad (S^t, \mathcal{R}^t, Y^t) \leftarrow \textsc{PeriodUpdate}(S^{t-1}, t, \vartheta)$
4: $\quad \mathcal{M}_{t\cdot} \leftarrow Y^t$
5: **return** $(S^T, \mathcal{M})$

3 Learning and Prediction

Let $\mathsf{L} \in \{\mathrm{NN}, \mathrm{Ridge}\}$ select the learning function. The default is $\mathsf{L} = \mathrm{NN}$. Both branches use the same histories, period window, and observation cap. Inputs remain on their raw DGP scale.

For any one-hidden-layer predictor with input dimension p_{in} and hidden width h,

$$\mathrm{NN}_\Theta(\xi) = w_2^\top \,\mathrm{ReLU}(W_1\xi + b_1) + b_2, \qquad \Theta = (W_1, b_1, w_2, b_2),$$

where $\xi \in \mathbb{R}^{p_{\text{in}}}$, $W_1 \in \mathbb{R}^{h\times p_{\text{in}}}$, $b_1, w_2 \in \mathbb{R}^h$, and $b_2 \in \mathbb{R}$.

$$d_b = d + \frac{d(d+1)}{2}, \qquad \psi_b(x_i, x_j) = \left[x_i + x_j;\ \mathrm{vech}\left(\frac{x_i x_j^\top + x_j x_i^\top}{2}\right)\right].$$

Agent networks use $(p_{\text{in}}, h) = (d, 2d)$; the broker network uses $(p_{\text{in}}, h) = (d_b, 8d)$.

$$\psi_b(x_i, x_j) = \psi_b(x_j, x_i)$$

so broker histories produce one training column per observed unordered pair.

Local variables for training

Symbol	Dimension/domain	Definition
p_{in}, h	integers	Predictor input dimension and hidden width.
ξ	$\mathbb{R}^{p_{\text{in}}}$	Predictor input vector.
L	learning function	NN or Ridge.
a_b	broker Ridge variant	Pair, size-matched, single-principal, or additive.
Θ	predictor parameters	NN parameters (W_1, b_1, w_2, b_2) or Ridge parameters (α, β).
W_1, b_1, w_2, b_2	matrix, vectors, scalar	One-hidden-layer MLP weights and biases.
d_b	integer	Broker symmetric pair-feature dimension.
ψ_b	function	Symmetric broker pair-feature map.
$\Xi_{\text{tr}}, y_{\text{tr}}$	$\mathbb{R}^{p_{\text{in}}\times n}$, $\mathbb{R}^n$	Generic training inputs and targets.
$\mathcal{L}$	scalar	Full-batch MSE objective.
λ	positive scalar	Learner-specific Ridge slope penalty; intercepts are not penalized.
e	integer	Update-step index.
(m, v, τ)	Adam state	Persistent Adam moments and step counter; $\mathcal{O}_i, \mathcal{O}_b$ are the agent and broker instances.
$\beta_1, \beta_2, \epsilon$	scalars	Adam constants (0.9, 0.999, 10^{-8}).
n_i, n_b	integers	Current values of history lengths n_i^t and n_b^t.
L_i, L_b	ordered index sets	Training-window indices from the most recent W_p completed simulation periods, with period 0 included only until W_p later periods exist; subsampled evenly to at most M.
Δ_i, Δ_b	integers	New observations since last training call.
E_i, E_b	integers	Agent and broker NN update-step counts.
$\Xi_{i,\text{tr}}, y_{i,\text{tr}}$	$\mathbb{R}^{d\times\lvert L_i\rvert}$, $\mathbb{R}^{\lvert L_i\rvert}$	Agent i training view of $\mathcal{H}_i^t$.
$Z_{b,\text{tr}}, y_{b,\text{tr}}$	matrix, $\mathbb{R}^{\lvert L_b\rvert}$	Broker fitting view using the selected input map.
r_ℓ	$\{1, 2\}$	Endpoint retained once when broker observation ℓ is recorded for the single-principal ablation.

Algorithm 2 TrainNN

Require: Network parameters Θ, inputs $\Xi_{\text{tr}} \in \mathbb{R}^{p_{\text{in}} \times n}$, targets $y_{\text{tr}} \in \mathbb{R}^n$, steps E, learning rate η_{lr}, persistent Adam state (m, v, τ).

for $e = 1, \ldots, E$ **do**

$\quad g \leftarrow \nabla_\Theta \mathcal{L}(\Theta), \quad \mathcal{L}(\Theta) = n^{-1} \sum_{\ell=1}^{n} \left(\mathrm{NN}_\Theta(\Xi_{\text{tr},\cdot\ell}) - y_{\text{tr},\ell}\right)^2$

$\quad \tau \leftarrow \tau + 1$

$\quad m \leftarrow \beta_1 m + (1 - \beta_1) g, \quad v \leftarrow \beta_2 v + (1 - \beta_2)\, g \odot g$

$\quad \Theta \leftarrow \Theta - \eta_{\text{lr}} \dfrac{m/(1-\beta_1^\tau)}{\sqrt{v/(1-\beta_2^\tau)} + \epsilon}$

return $\Theta, (m, v, \tau)$

Algorithm 3 FitPredictor

Require: Learning function L, learner $c \in \{a, b\}$, parameters Θ, raw inputs $\Xi_{\text{tr}} \in \mathbb{R}^{p_{\text{in}} \times n}$, targets $y_{\text{tr}} \in \mathbb{R}^n$, learner-specific Ridge penalty λ_c, and fitting parameters.

if $\mathsf{L} = \mathrm{NN}$ **then**

$\quad E \leftarrow \max\{E_{\text{steps},c}, \lceil E_{\text{init},c} \Delta / n_{\text{total}} \rceil\}$.

$\quad (\Theta, \mathcal{O}) \leftarrow \textsc{TrainNN}(\Theta, \Xi_{\text{tr}}, y_{\text{tr}}, E, \eta_{\text{lr},c}, \mathcal{O})$.

else

$\quad (\alpha, \beta) \leftarrow \arg\min_{a,b} \left\{ n^{-1} \sum_{\ell=1}^{n} (a + b^\top \Xi_{\text{tr},\cdot\ell} - y_{\text{tr},\ell})^2 + \lambda_c \|b\|_2^2 \right\}$.

$\quad \Theta \leftarrow (\alpha, \beta)$.

return Θ and, for NN, $\mathcal{O}$.

The agent input is always the prospective partner type. The broker Ridge training input for observation ℓ is

$$z_{b\ell} = \begin{cases} \psi_b(x_{1\ell}, x_{2\ell}), & a_b \in \{\text{pair}, \text{size-matched}\}, \\ x_{r_\ell,\ell}, & a_b = \text{single-principal}, \\ x_{1\ell} + x_{2\ell}, & a_b = \text{additive}. \end{cases}$$

For size-matched history, uniformly sample without replacement from the broker's windowed and capped observations. Its sample size is the smaller of the broker sample size and the nearest-integer median agent sample size among agents with nonempty histories, with half-integers rounded up.

Prediction is modular:

$$\textsc{PredictAgent}(i, x_j) = \begin{cases} \mathrm{NN}_{\Theta_i}(x_j), & \mathsf{L} = \mathrm{NN}, \\ \alpha_i + \beta_i^\top x_j, & \mathsf{L} = \mathrm{Ridge}, \end{cases}$$

and

$$\textsc{PredictBroker}(i, j) = \begin{cases} \mathrm{NN}_{\Theta_b}(\psi_b(x_i, x_j)), & \mathsf{L} = \mathrm{NN}, \\ \alpha_b + \gamma_b^\top \psi_b(x_i, x_j), & \mathsf{L} = \mathrm{Ridge},\ a_b \in \{\text{pair}, \text{size-matched}\}, \\ \alpha_b + \gamma_b^\top (x_i + x_j), & \mathsf{L} = \mathrm{Ridge},\ a_b = \text{additive}, \\ g_b(x_i) + g_b(x_j) - \bar{y}_{b,\text{tr}}, & \mathsf{L} = \mathrm{Ridge},\ a_b = \text{single-principal}, \end{cases}$$

where $g_b(x) = \alpha_b + \gamma_b^\top x$. Before any Ridge fit, $\alpha = Q$, slopes are zero, and $\bar{y}_{b,\text{tr}} = Q$.

Within period-level routines, state variables without a superscript refer to their current-period values.

Algorithm 4 TrainPredictors

Require: State S, period t, learning function L, and fitting parameters.

for $i = 1, \dots, N$ in parallel **do**

$n_i \leftarrow n_i^t$, $\Delta_i \leftarrow$ observations added to $\mathcal{H}_i$ since its last training call.

if $n_i > 0$, $\Delta_i > 0$, and $i \bmod 2 = t \bmod 2$ **then**

Let $P_i \leftarrow |\mathbf{p}_i|$ and $b_i \leftarrow 0$ if $P_i \le W_p$, otherwise $b_i \leftarrow (\mathbf{p}_i)_{P_i - W_p}$; set $L_i \leftarrow \{b_i + 1, \dots, n_i\}$. If $|L_i| > M$, keep M spaced indices across L_i, including its first and last indices (or only its last index when $M = 1$).

$\Xi_{i,\mathrm{tr}} \leftarrow (\Xi_i^t)_{\cdot L_i}$, $y_{i,\mathrm{tr}} \leftarrow (y_i^t)_{L_i}$.

$\Theta_i \leftarrow \textsc{FitPredictor}(\mathsf{L}, a, \Theta_i, \Xi_{i,\mathrm{tr}}, y_{i,\mathrm{tr}}, \lambda_a, \Delta_i, n_i, \vartheta)$; $\Delta_i \leftarrow 0$.

$n_b \leftarrow n_b^t$, $\Delta_b \leftarrow$ broker observations since last training.

if $n_b > 0$ and $\Delta_b > 0$ **then**

Let $P_b \leftarrow |\mathbf{p}_b|$ and $b_b \leftarrow 0$ if $P_b \le W_p$, otherwise $b_b \leftarrow (\mathbf{p}_b)_{P_b - W_p}$; set $L_b \leftarrow \{b_b + 1, \dots, n_b\}$. If $|L_b| > M$, keep M spaced indices across L_b, including its first and last indices (or only its last index when $M = 1$).

If $\mathsf{L} =$ Ridge and $a_b =$ size-matched, replace L_b by the uniform size-matched sample defined above.

Construct $Z_{b,\mathrm{tr}} = [z_{b\ell}]_{\ell \in L_b}$ and $y_{b,\mathrm{tr}} \leftarrow (y_b^t)_{L_b}$.

$\Theta_b \leftarrow \textsc{FitPredictor}(\mathsf{L}, b, \Theta_b, Z_{b,\mathrm{tr}}, y_{b,\mathrm{tr}}, \lambda_b, \Delta_b, n_b, \vartheta)$; $\Delta_b \leftarrow 0$.

4 Initialization

Calibration draws random pairs (i_ℓ, j_ℓ) and sets

$$\bar{q}_{\mathrm{cal}} = \frac{1}{10{,}000} \sum_{\ell=1}^{10{,}000} \{Q + f(x_{i_\ell}, x_{j_\ell})\},$$
$$r = f_r\, \bar{q}_{\mathrm{cal}}, \qquad \phi = c_s = \lambda_c\, \bar{q}_{\mathrm{cal}}.$$

Local variables for Initialize

Symbol	Dimension/domain	Definition
λ_c	scalar	Cost rate used to set ϕ and c_s.
f_r	scalar	Reservation coefficient; sets $r = f_r\, \bar{q}_{\mathrm{cal}}$ (default 0.60).
(i_ℓ, j_ℓ)	pair in $[N]^2$	Random calibration pair.
d_γ	integer	Number of active curve dimensions.
p_{rewire}	scalar	Watts-Strogatz rewiring probability.
ν_ℓ	integer	Frequency for active curve coordinate ℓ.
ψ_ℓ	scalar	Phase for active curve coordinate ℓ.
A_0	$\mathbb{R}^{d \times d}$	Raw matrix used to construct A.
Σ_x	$\mathbb{R}^{d \times d}$	Empirical agent type second moment.
B_0	$\mathbb{R}^{d \times d}$	Raw zero-trace symmetric boundary matrix.
B_{raw}	$\mathbb{R}^{d \times d}$	Orthogonalized pre-normalization boundary matrix.
$Z_{b,\mathrm{seed}}, y_{b,\mathrm{seed}}$	matrix, $\mathbb{R}^{n_b^0}$	Broker seed fitting batch using the selected input map.

Local variables for InitializeAgent

Symbol	Dimension/domain	Definition
σ_x	scalar	Agent type noise scale.
u_i	scalar	Agent i curve position.
ϵ_i	$\mathbb{R}^d$	Agent type noise for agent i.
π_{ij}^{ent}	scalar	Entrant attachment probability.
$\mathcal{V}_i^{\text{ent}}$	subset of $[N] \setminus \{i\}$	Entrant attachment set.

Algorithm 5 InitializeAgent

Require: State S, agent slot i, period counter t, parameters ϑ.
Ensure: Agent slot i is initialized in S.

Draw $u_i \sim \text{Unif}[0,1]$ and $\epsilon_i \sim \mathcal{N}(0, \sigma_x^2 I_d/d)$.
$x_i^t \leftarrow (\gamma(u_i) + \epsilon_i)/\|\gamma(u_i) + \epsilon_i\|_2$.
Reset $\mathcal{H}_i^t = (\Xi_i^t, y_i^t, n_i^t)$ to zero observations, $\bar{q}_{i\cdot}^t$, $C_{i\cdot}^t$, χ_i^t, M_i^t, and Δ_i^t.
Initialize the selected predictor to the constant Q: for NN, draw W_{1i}^t with He initialization and set $b_{1i}^t = 0$, $w_{2i}^t = 0$, $b_{2i}^t = Q$, $\mathcal{O}_i^t = 0$; for Ridge, set $\alpha_i^t = Q$ and $\beta_i^t = 0$.
if $t = 0$ **then**
  $s_i^t \leftarrow (0, 0)$.
else
  Remove i from Roster^t, D^t, and M_j^t for all j; delete all incident edges in G^t; set $\bar{q}_{ji}^t, C_{ji}^t \leftarrow 0$ for all $j \neq i$.
  Sample $\mathcal{V}_i^{\text{ent}} \subset [N] \setminus \{i\}$ without replacement, $|\mathcal{V}_i^{\text{ent}}| = \min(\lfloor k_G/2 \rfloor, N-1)$, with

$$\pi_{ij}^{\text{ent}} \propto \exp\{-\|x_i^t - x_j^t\|_2^2\}.$$

  Add edges (i,j) to G^t for all $j \in \mathcal{V}_i^{\text{ent}}$.

$$s_{i,\text{self}}^t \leftarrow \begin{cases} |\mathcal{V}_i^{\text{ent}}|^{-1} \sum_{j \in \mathcal{V}_i^{\text{ent}}} s_{j,\text{self}}^t, & \mathcal{V}_i^{\text{ent}} \neq \emptyset, \\ 0, & \mathcal{V}_i^{\text{ent}} = \emptyset. \end{cases}$$

  $s_{i,\text{broker}}^t \leftarrow \text{rep}_b^t$.
  Set $\mathbf{p}_i^t \leftarrow (0)$, treating entrant initialization as period 0 of its tenure.

Algorithm 6 Initialize

Require: Parameters ϑ and seed.
Ensure: Initial state S^0.

Draw curve frequencies $\nu_\ell \sim \text{Unif}\{1, \ldots, 5\}$ and phases $\psi_\ell \sim \text{Unif}[0, 2\pi)$ for $\ell = 1, \ldots, d_\gamma$, defining γ.
Allocate empty agent and broker containers in S^0 and store γ.
for $i = 1, \ldots, N$ **do**
  $\textsc{InitializeAgent}(S^0, i, 0, \vartheta)$.
Draw $u_c \sim \text{Unif}[0,1]$ and $\epsilon_c \sim \mathcal{N}(0, \sigma_x^2 I_d/d)$; set $c \leftarrow [\gamma(u_c) + \epsilon_c]/\|\gamma(u_c) + \epsilon_c\|_2$.
Draw $A_0 \in \mathbb{R}^{d \times d}$ iid standard normal and set $A \leftarrow A_0^\top A_0 \cdot d / \text{tr}(A_0^\top A_0)$.
$\Sigma_x \leftarrow N^{-1} \sum_i x_i^0 (x_i^0)^\top$.
Draw symmetric zero-trace B_0 and set

$$B_{\text{raw}} \leftarrow B_0 - \frac{\text{tr}(\Sigma_x B_0 \Sigma_x A)}{\text{tr}(\Sigma_x A \Sigma_x A)} A, \qquad B \leftarrow B_{\text{raw}}/\|B_{\text{raw}}\|_F.$$

Calibrate $(\bar{q}_{\mathrm{cal}}, r, \phi, c_s)$.
Build G^0 as a Watts-Strogatz graph on $[N]$ with degree k_G and rewiring probability p_{rewire}; add isolated broker node $N+1$.
Initialize $\mathcal{H}_b^0$ and the selected broker predictor to the constant Q using the corresponding NN or Ridge initialization above; set $\mathrm{rep}_b^0 = 0$.
Sample $\mathrm{Roster}^0 \subset [N]$ uniformly with $|\mathrm{Roster}^0| = \lceil \alpha_R N \rceil$; add broker edges $(i, N+1)$ for all roster members.
for each unordered non-broker edge $\{i,j\} \in \mathcal{E}_G^0$ **do**
    Draw q_{ij} once; append (x_j^0, q_{ij}) to $\mathcal{H}_i^0$ and (x_i^0, q_{ij}) to $\mathcal{H}_j^0$; update $\bar{q}_{ij}^0, C_{ij}^0, \bar{q}_{ji}^0, C_{ji}^0$.
Let $\mathcal{B}_{\mathrm{seed}} \leftarrow \{\{i,j\} \in \mathcal{E}_G^0 : i,j \in \mathrm{Roster}^0\}$; shuffle and keep at most 100 edges.
for each $\{i,j\} \in \mathcal{B}_{\mathrm{seed}}$ **do**
    Append (x_i^0, x_j^0, q_{ij}) to $\mathcal{H}_b^0$; increment n_b^0.
    If $\mathsf{L} = \mathrm{Ridge}$ and $a_b = \mathrm{single\text{-}principal}$, draw and store $r_\ell \sim \mathrm{Unif}\{1,2\}$ for this observation.
if $n_b^0 > 0$ **then**
    Set rep_b^0 to mean broker seed output.
else
    Set $\mathrm{rep}_b^0 \leftarrow 0$.
Set $s_{i,\mathrm{self}}^0$ to mean own seed output and $s_{i,\mathrm{broker}}^0 \leftarrow \mathrm{rep}_b^0$.
Set $\mathbf{p}_i^0 \leftarrow (n_i^0)$ for every agent and $\mathbf{p}_b^0 \leftarrow (n_b^0)$, so initialization is period 0; set $\Delta_i^0 \leftarrow n_i^0$ and $\Delta_b^0 \leftarrow n_b^0$.
for each i with $n_i^0 > 0$ **do**
    $\Theta_i^0 \leftarrow \textsc{FitPredictor}(\mathsf{L}, a, \Theta_i^0, \Xi_i^0, y_i^0, \lambda_a, n_i^0, n_i^0, \vartheta)$; $\Delta_i^0 \leftarrow 0$.
if $n_b^0 > 0$ **then**
    $Z_{b,\mathrm{seed}} \leftarrow [z_{b\ell}]_{\ell=1}^{n_b^0}$, $y_{b,\mathrm{seed}} \leftarrow y_b^0$.
    $\Theta_b^0 \leftarrow \textsc{FitPredictor}(\mathsf{L}, b, \Theta_b^0, Z_{b,\mathrm{seed}}, y_{b,\mathrm{seed}}, \lambda_b, n_b^0, n_b^0, \vartheta)$; $\Delta_b^0 \leftarrow 0$.
return S^0.

5 Channel Choice

Local variables for ChooseChannel

Symbol	Dimension/domain	Definition
U_{self}	scalar	Self-channel score.
U_{broker}	scalar	Broker-channel score.

Algorithm 7 ChooseChannel

Require: Agent i state (s_i, χ_i), broker reputation rep_b, RNG.
$U_{\mathrm{self}} \leftarrow s_{i,\mathrm{self}}$.
$U_{\mathrm{broker}} \leftarrow \chi_i s_{i,\mathrm{broker}} + (1-\chi_i)\mathrm{rep}_b$.
if $U_{\mathrm{self}} > U_{\mathrm{broker}}$ **then**
    return self.
if $U_{\mathrm{broker}} > U_{\mathrm{self}}$ **then**
    return broker.
return self w.p. 1/2, else broker ▷ $U_{\mathrm{self}} = U_{\mathrm{broker}}$.

6 One Period

Local variables for PeriodUpdate

Symbol	Dimension/domain	Definition
K	integer	Active-demand bound.
p_{demand}	scalar	Per-position demand probability.
α_R	scalar	Standing broker roster share.
p_{roster}	scalar	Roster-churn probability.
η_{exit}	scalar	Agent exit probability.
Δ_{net}	integer	Network-measure interval.

Algorithm 8 PeriodUpdate

Require: Previous state S^{t-1}, period t, parameters ϑ.
Ensure: Updated state S^t, accepted records $\mathcal{R}^t$, and pre-turnover measure row Y^t.

Clear accumulators, set $M_i^t \leftarrow \emptyset$ for all i, and set $D^t \leftarrow \emptyset$.
Refresh roster: remove each roster member with probability p_{roster}, then replenish uniformly to size $\lceil \alpha_R N \rceil$.
$\mathcal{E}_G^t \leftarrow (\mathcal{E}_G^t \setminus \{\{i, N+1\} : i \in [N]\}) \cup \{\{i, N+1\} : i \in \text{Roster}^t\}$.
$\mathcal{D}^t \leftarrow \emptyset$.
for $i = 1, \ldots, N$ **do**
  Draw $d_i^t \sim \text{Binomial}(K, p_{\text{demand}})$.
  if $d_i^t > 0$ **then**
    $z_i^t \leftarrow \textsc{ChooseChannel}(i, \text{rep}_b^{t-1})$.
    Append (i, z_i^t, d_i^t) to $\mathcal{D}^t$.
    if $z_i^t = \text{broker}$ **then**
      Add i to D^t.
$\mathcal{E}_G^t \leftarrow (\mathcal{E}_G^t \setminus \{\{i, N+1\} : i \in [N]\}) \cup \{\{i, N+1\} : i \in \text{Roster}^t \cup D^t\}$.
$\textsc{TrainPredictors}(S, t, \vartheta)$.
$\mathcal{R}^t \leftarrow \textsc{OfferMarket}(S, \mathcal{D}^t, \vartheta)$.
Synchronize broker-node edges to Roster^t, D^t, and current broker-channel matches.
$\textsc{UpdateSatisfaction}(S, \mathcal{D}^t, \mathcal{R}^t)$.
if $D^t \neq \emptyset$ **then**
  $\text{rep}_b^t \leftarrow |D^t|^{-1} \sum_{i \in D^t} s_{i,\text{broker}}^t$.
else
  $\text{rep}_b^t \leftarrow \text{rep}_b^{t-1}$.
if $t \bmod \Delta_{\text{net}} = 0$ **then**
  Recompute broker betweenness, Burt constraint, and effective size on the post-matching, pre-turnover graph G^t.
$Y^t \leftarrow \textsc{CollectMeasures}(S^t, \mathcal{R}^t, \vartheta)$ on the post-matching, pre-turnover state.
Append n_i^t to $\mathbf{p}_i^t$ for every agent and append n_b^t to $\mathbf{p}_b^t$, recording the end-of-period training-window boundaries.
for $i = 1, \ldots, N$ **do**
  if agent i exits with probability η_{exit} **then**
    $\textsc{InitializeAgent}(S^t, i, t, \vartheta)$.
Synchronize broker-node edges to the post-turnover roster, clients, and active matches.
return $(S^t, \mathcal{R}^t, Y^t)$.

7 Offer Market

Local variables for OfferMarket

Symbol	Dimension/domain	Definition
$\hat{q}_i^t$	$\mathbb{R}^d \to \mathbb{R}$	Agent predictor.
$\hat{q}_b^t$	$\mathbb{R}^{d_b} \to \mathbb{R}$	Broker predictor.
$\mathcal{N}_G^t(i)$	subset of $[N]$	Non-broker graph neighbors of i.
U_i^t	subset of $[N]$	Self-search stranger pool independently sampled for demander i.
$n_{\text{strangers}}$	integer	Per-demander stranger-pool size.
$v_{i\to j}^{\text{self},t}$	scalar	Self-offer score.
$\mathcal{U}_{b,i}$	subset of $[N]$	Candidate set for broker-client demander i.
$v_{b,i\to j}^t$	scalar	Score used to order brokered candidates for demander i.
Ω_i	integer	Residual offer quota for demander i.
Ω	$\mathbb{Z}_{\geq 0}^N$	Residual quota vector.
O^t	map	Directed offer book.
O_{ij}^t	offer record	Record (i, j, z, v) for offer $i \to j$.
$\mathcal{P}_O^t$	set	Unordered pairs represented in O^t.
$\mathcal{U}_i^{\text{self}}$	subset of $[N]$	Self-search candidate set.
$\mathcal{I}_i^{\text{self}}$	list	Self-search candidates sorted by score.
$\mathcal{I}_{b,i}$	list	Brokered candidates for client i, sorted by $v_{b,i\to j}^t$.
$\mathcal{B}_b^t$	list	Selected directed broker offers.
$u_{i\to j}^{\text{self}}, u_{b,i\to j}^{\text{local}}, u_{b,i\to j}^{\text{global}}$	$[0,1]$	Independent random secondary keys used only within equal-score rankings.
o, o', o_{ij}, o_{ji}	offer records	Offer records; fields include $o.z$ and o.from.
$z^\star$	channel	Accepted-record channel.
ι	$\{0,1\}$	Return value of AddOffer.

The symmetric broker score is

$$v_b^t(i,j) = \textsc{PredictBroker}(i,j).$$

Let $\mathcal{N}_G^t(i) = \{j \in [N] : (i,j) \in \mathcal{E}_G^t\}$ exclude the broker node. Agent i assesses any presented candidate as

$$a_{i\to j}^t = \begin{cases} \bar{q}_{ij}^t, & j \in \mathcal{N}_G^t(i),\ C_{ij}^t > 0, \\ \textsc{PredictAgent}(i, x_j^t), & \text{otherwise.} \end{cases}$$

Self-search makes this score available only for neighbors and non-neighbors in the demander-specific stranger pool:

$$v_{i\to j}^{\text{self},t} = \begin{cases} a_{i\to j}^t, & j \in \mathcal{N}_G^t(i) \cup (U_i^t \setminus \mathcal{N}_G^t(i)), \\ \perp, & \text{otherwise.} \end{cases}$$

Thus direct pair means replace predictor extrapolations when pair-specific experience exists; $\perp$ denotes no valid self-search score. The receiver-side acceptance score for a one-sided offer is

$$w_{j\leftarrow i}^t = \begin{cases} \bar{q}_{ji}^t, & (i,j) \in \mathcal{E}_G^t,\ C_{ji}^t > 0, \\ \textsc{PredictAgent}(j, x_i^t), & \text{otherwise.} \end{cases}$$

The market creates at most d_i^t outgoing offers from demander i. Receivers have no per-period capacity constraint; acceptance is pairwise, so a receiver may accept multiple offers in the same period. A demanded position can remain unmatched if no acceptable offer is formed.

Algorithm 9 AddOffer

Require: Offer book O^t, unordered-pair set $\mathcal{P}_O^t$, offer (i, j, z, v).

if O_{ij}^t is defined **then**
  return 0.
$O_{ij}^t \leftarrow (i, j, z, v)$.
if O_{ji}^t is undefined **then**
  $\mathcal{P}_O^t \leftarrow \mathcal{P}_O^t \cup \{\{i, j\}\}$.
return 1.

Algorithm 10 BrokerCandidates

Require: Broker-client demander i, state S, broker service S_b.
Ensure: Candidate set $\mathcal{U}_{b,i}$ and score function $v_{b,i\to j}^t$.

if $\mathsf{S}_b =$ full **then**
  $\mathcal{U}_{b,i} \leftarrow \mathcal{A}_b^t \setminus (\{i\} \cup M_i^t)$ and $v_{b,i\to j}^t \leftarrow v_b^t(i,j)$.
else if $\mathsf{S}_b =$ assessment-only **then**
  Draw $U_i^t \subset [N]$ uniformly without replacement, $|U_i^t| = \min(n_{\text{strangers}}, N)$.
  $\mathcal{U}_{b,i} \leftarrow [\mathcal{N}_G^t(i) \cup (U_i^t \setminus \mathcal{N}_G^t(i))] \setminus (\{i\} \cup M_i^t)$ and $v_{b,i\to j}^t \leftarrow v_b^t(i,j)$.
else
  $\mathcal{U}_{b,i} \leftarrow \mathcal{A}_b^t \setminus (\{i\} \cup M_i^t)$ and $v_{b,i\to j}^t \leftarrow a_{i\to j}^t$. ▷ access-only
return $(\mathcal{U}_{b,i}, v_{b,i\to j}^t)$.

Algorithm 11 OfferMarket

Require: State S, demand records $\mathcal{D}^t$, parameters ϑ.
Ensure: Accepted records $\mathcal{R}^t$.

For all $(i, z_i, d_i) \in \mathcal{D}^t$, set residual offer quota $\Omega_i \leftarrow d_i$.
$O^t \leftarrow \emptyset$, $\mathcal{P}_O^t \leftarrow \emptyset$, $\mathcal{R}^t \leftarrow \emptyset$.
for each $(i, \text{self}, d_i) \in \mathcal{D}^t$ with $\Omega_i > 0$ **do**
  Draw $U_i^t \subset [N]$ uniformly without replacement, $|U_i^t| = \min(n_{\text{strangers}}, N)$.
  $\mathcal{U}_i^{\text{self}} \leftarrow \mathcal{N}_G^t(i) \cup (U_i^t \setminus \mathcal{N}_G^t(i))$.
  Remove from $\mathcal{U}_i^{\text{self}}$ the elements of $\{i\} \cup M_i^t$.
  Draw independent $u_{i\to j}^{\text{self}} \sim \text{Unif}(0,1)$ for each feasible j.
  $\mathcal{I}_i^{\text{self}} \leftarrow \operatorname{argsort}_{j \in \mathcal{U}_i^{\text{self}}:\ v_{i\to j}^{\text{self},t} > r}(-v_{i\to j}^{\text{self},t}, u_{i\to j}^{\text{self}})$.
  for the first $\min(\Omega_i, |\mathcal{I}_i^{\text{self}}|)$ agents j in $\mathcal{I}_i^{\text{self}}$ **do**
    $\textsc{AddOffer}(O^t, \mathcal{P}_O^t, i, j, \text{self}, v_{i\to j}^{\text{self},t})$.
$\mathcal{A}_b^t \leftarrow \text{Roster}^t \cup D^t$.
$D_+^t \leftarrow \{i : (i, \text{broker}, d_i) \in \mathcal{D}^t,\ \Omega_i > 0\}$.
$\mathcal{B}_b^t \leftarrow \emptyset$.
for each $i \in D_+^t$ **do**
  $(\mathcal{U}_{b,i}, v_{b,i\to j}^t) \leftarrow \textsc{BrokerCandidates}(i, S, \mathsf{S}_b)$.
  Draw independent $u_{b,i\to j}^{\text{local}} \sim \text{Unif}(0,1)$ for each feasible j.
  $\mathcal{I}_{b,i} \leftarrow \operatorname{argsort}_{j \in \mathcal{U}_{b,i}:\ v_{b,i\to j}^t > r}(-v_{b,i\to j}^t, u_{b,i\to j}^{\text{local}})$.

for the first $\min(\Omega_i, |\mathcal{I}_{b,i}|)$ agents j in $\mathcal{I}_{b,i}$ **do**
Append $(i, j, v^t_{b,i\to j})$ to $\mathcal{B}^t_b$.
Draw a new independent $u^{\text{global}}_{b,i\to j} \sim \text{Unif}(0,1)$ for each $(i, j, v) \in \mathcal{B}^t_b$.
Sort $\mathcal{B}^t_b$ lexicographically by $(-v, u^{\text{global}}_{b,i\to j})$.
for each $(i, j, v) \in \mathcal{B}^t_b$ **do**
if $\Omega_i > 0$ **then**
$\iota \leftarrow \textsc{AddOffer}(O^t, \mathcal{P}^t_O, i, j, \text{broker}, v)$.
$\Omega_i \leftarrow \Omega_i - \iota$.
for each $\{i, j\} \in \mathcal{P}^t_O$ **do**
$o_{ij} \leftarrow O^t_{ij}$ if defined, else $\emptyset$; $o_{ji} \leftarrow O^t_{ji}$ if defined, else $\emptyset$.
if $o_{ij} \neq \emptyset$ and $o_{ji} \neq \emptyset$ **then**
$\textsc{FinalizeMatch}(S, \mathcal{R}^t, o_{ij}, o_{ji})$.
else if $o_{ij} \neq \emptyset$ and $w^t_{j\leftarrow i} > r$ **then**
$\textsc{FinalizeMatch}(S, \mathcal{R}^t, o_{ij}, \emptyset)$.
else if $o_{ji} \neq \emptyset$ and $w^t_{i\leftarrow j} > r$ **then**
$\textsc{FinalizeMatch}(S, \mathcal{R}^t, o_{ji}, \emptyset)$.
return $\mathcal{R}^t$.

Algorithm 12 FinalizeMatch

Require: State S, accepted-record list $\mathcal{R}^t$, offer $o = (i, j, z, v)$, optional reverse offer o'.
if $j \in M^t_i$ **then**
return .
Draw $q^t_{ij} = Q + f(x^t_i, x^t_j) + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, \sigma^2_\varepsilon)$.
Append columns x^t_j to Ξ_i and x^t_i to Ξ_j; append q^t_{ij} to y_i and y_j; increment n_i and n_j.
Update $\bar{q}_{ij}, C_{ij}$ and $\bar{q}_{ji}, C_{ji}$.
Add edge (i, j) to G^t; add j to M^t_i and i to M^t_j.
if $z = \text{broker}$ or ($o' \neq \emptyset$ and $o'.z = \text{broker}$) **then**
Append the two party types x^t_i and x^t_j to $X_{b,1}$ and $X_{b,2}$; append q^t_{ij} to y_b; increment n_b.
If $\mathsf{L} = \text{Ridge}$ and $a_b = \text{single-principal}$, draw and store $r_\ell \sim \text{Unif}\{1, 2\}$ for this observation.
$z^\star \leftarrow \text{broker}$.
else
$z^\star \leftarrow \text{self}$.
Append accepted record $(i, j, z^\star, q^t_{ij}, v, o, o')$ to $\mathcal{R}^t$.

8 Satisfaction and Measures

Local variables for satisfaction

Symbol	Dimension/domain	Definition
Y_i	scalar	Agent i period satisfaction numerator.
$n_{i,b}$	integer	Accepted broker offers credited to demander i.
$\tilde{q}^t_i$	scalar	Channel-specific satisfaction input.
ζ	accepted record	Iterator over $\mathcal{R}^t$.
$\mathbf{1}\{\cdot\}$	indicator	Equals 1 when its condition is true and 0 otherwise.

Algorithm 13 UpdateSatisfaction

Require: State S, demand records $\mathcal{D}^t$, accepted records $\mathcal{R}^t$.

for each $(i, z_i, d_i) \in \mathcal{D}^t$ **do**
    $Y_i \leftarrow \sum_{\zeta \in \mathcal{R}^t} \sum_{o \in \{\zeta.o_1, \zeta.o_2\}} \mathbf{1}\{o \neq \emptyset,\ o.\mathrm{from} = i\}\, \zeta.q$.
    $n_{i,b} \leftarrow \sum_{\zeta \in \mathcal{R}^t} \sum_{o \in \{\zeta.o_1, \zeta.o_2\}} \mathbf{1}\{o \neq \emptyset,\ o.\mathrm{from} = i,\ o.z = \mathrm{broker}\}$.
    if $z_i = \mathrm{self}$ **then**
        $\tilde{q}_i^t \leftarrow Y_i / d_i - c_s$.
    else
        $\tilde{q}_i^t \leftarrow (Y_i - \phi n_{i,b}) / d_i$; $\chi_i^t \leftarrow 1$.
    $s_{i,z_i}^t \leftarrow (1 - \omega) s_{i,z_i}^{t-1} + \omega \tilde{q}_i^t$.

CollectMeasures. $\textsc{CollectMeasures}(S^t, \mathcal{R}^t, \vartheta)$ records the measures defined in Appendix B, Section 8, without changing the simulation state.